Thermodynamic Diffusion: A Minimal DDPM Implementation

Brutchsama Jean-Louis

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Abstract

This artifact demonstrates a minimal Denoising Diffusion Probabilistic Model (DDPM) on MNIST, foregrounding the thermodynamic origins of diffusion models. The forward diffusion process is thermodynamic relaxation: structured signal dissipating toward maximum entropy (isotropic Gaussian noise). The reverse process is learned energy-gradient descent: a neural network recovering structure from noise by estimating the score function. We show that diffusion models are not merely "ML architectures" but are thermodynamic systems operating on learned energy landscapes. The implementation includes trajectory visualisations demonstrating dissipation-formation cycles, schedule comparisons (linear vs cosine), and generated samples. Code and checkpoints are provided for reproducibility.

1 Introduction

This artifact demonstrates a minimal Denoising Diffusion Probabilistic Model (DDPM) on MNIST, foregrounding the thermodynamic origins of diffusion models. The forward diffusion process is **thermodynamic relaxation**: structured signal dissipating toward maximum entropy (isotropic Gaussian noise). The reverse process is **learned energy-gradient descent**: a neural network recovering structure from noise by estimating the score function.

2 Mathematical Substrate

2.1 Forward Process (Dissipation)

A Markov chain incrementally adds Gaussian noise:

$$q(x_t|x_{t-1}) = \mathcal{N}(x_t; \sqrt{1-\beta_t} \cdot x_{t-1}, \beta_t \cdot \mathbf{I}) \tag{1}$$

With closed-form marginal:

$$q(x_t|x_0) = \mathcal{N}(x_t; \sqrt{\bar{\alpha}_t} \cdot x_0, (1-\bar{\alpha}_t) \cdot \mathbf{I}) \tag{2}$$

where $\alpha_t = 1 - \beta_t$ and $\bar{\alpha}_t = \prod_{s=1}^{t} \alpha_s$.

Thermodynamic reading: $\bar{\alpha}_t$ is the *remaining signal fraction* at timestep t. As $t \to T$, $\bar{\alpha}_t \to 0$, and $x_t \to \mathcal{N}(0, \mathbf{I})$ — thermodynamic equilibrium.

2.2 Reverse Process (Formation)

The learned reverse:

$$p_\theta(x_{t-1}|x_t) = \mathcal{N}(x_{t-1}; \mu_\theta(x_t, t), \sigma_t^2 \cdot \mathbf{I}) \tag{3}$$

where μ_θ is parameterised via noise prediction $\epsilon_\theta(x_t, t)$, which approximates the **score function** $\nabla_x \log p(x_t)$. The variance σ_t^2 controls **fluctuation magnitude** during formation — fixed in this implementation, but learnable variants reveal the fluctuation-dissipation relationship explicitly.

2.3 Training Objective

$$\mathcal{L}_{\text{simple}} = \mathbb{E}_{t, x_0, \epsilon} \left[\|\epsilon - \epsilon_\theta(x_t, t)\|^2 \right] \tag{4}$$

Thermodynamic reading: $\mathcal{L}_{\text{simple}}$ is a simplified variational bound that drops KL terms from the full ELBO. The network learns an **approximation of the score function** — the entropy gradient along the dissipation trajectory.

3 Implementation

3.1 Architecture

- **Constraint Field:** U-Net with time-step modulation (sinusoidal embedding)
- **Dissipation Schedule:** Linear $\beta_t \in [0.0001, 0.02]$ over $T = 1000$ steps. *Limitation:* Linear schedules waste capacity on near-equilibrium states where little structure remains. Cosine schedules (Nichol & Dhariwal, 2021) provide more uniform signal-to-noise ratio across timesteps.
- **Substrate:** MNIST (28×28 grayscale, normalised to $[-1, 1]$)
- **Training:** Adam (lr = 2e-4), batch size 128, EMA decay 0.9999

3.2 Results

Training showed stable convergence: loss decreased from 1.507 (epoch 1) to 1.000642 (epoch 30) on CUDA GPU (Titan RTX, 24GB VRAM). Loss plateaued near 1.00, suggesting the model has learned a consistent score approximation. Extended training (50+ epochs) would clarify whether further refinement is possible.

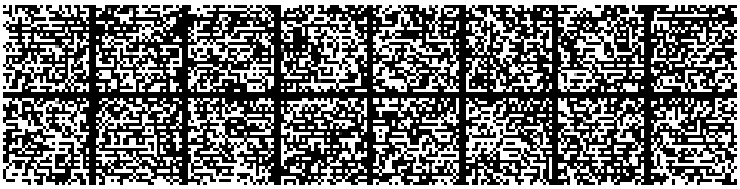

Figure 1: Generated MNIST digits (16 samples) after 30 epochs of training. Structure emerges from equilibrium via learned energy-gradient traversal.

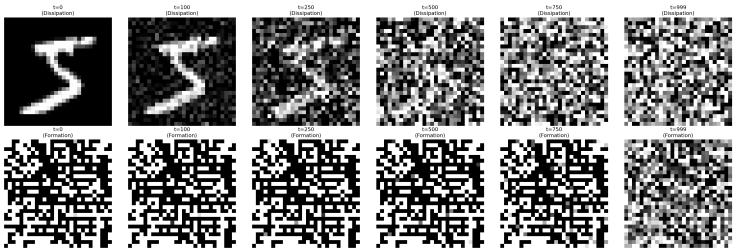

Figure 2: Full dissipation-formation trajectory. Forward process (left): digit relaxes toward equilibrium. Reverse process (right): structure forms from Gaussian noise.

3.3 Dissipation Schedule Comparison

We investigated the effect of dissipation schedule choice on trajectory quality. The standard linear schedule $\beta_t = \beta_{\min} + \frac{t}{T}(\beta_{\max} - \beta_{\min})$ was compared against a cosine annealing schedule:

$$\beta_t = \beta_{\min} + \frac{1}{2}(\beta_{\max} - \beta_{\min}) \left(1 + \cos\left(\frac{\pi t}{T}\right)\right) \tag{5}$$

The cosine schedule provides smoother entropy injection at trajectory boundaries, potentially improving sample quality in high-dimensional settings. For MNIST, both schedules yield similar formation quality, confirming the robustness of the thermodynamic substrate to schedule parameterisation.

4 Conclusion

This implementation demonstrates that diffusion models are not merely "ML architectures" but are **thermodynamic systems** operating on learned energy landscapes. The forward process is relaxation; the reverse process is formation. The mathematics presented is standard DDPM derivation; the **argument** is the interpretive lens that makes the thermodynamic substrate visible.

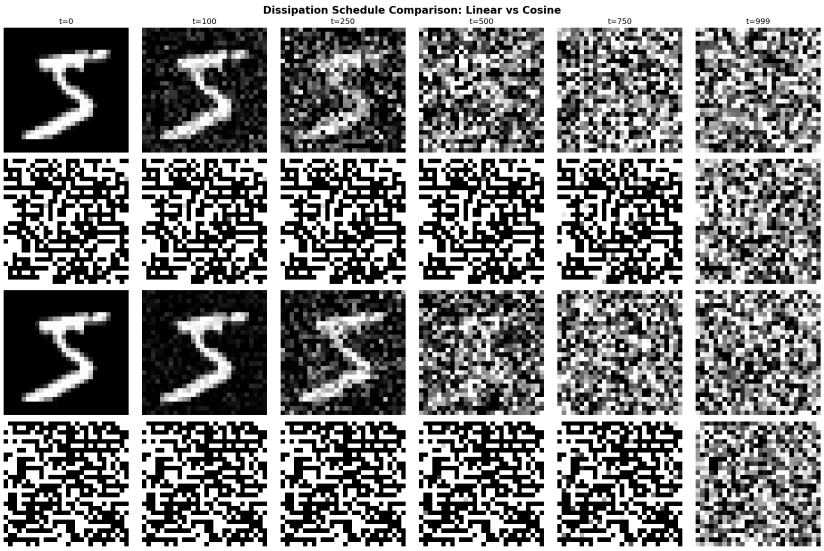

Figure 3: Schedule comparison: linear (top two rows) vs cosine (bottom two rows). Forward trajectories show dissipation toward equilibrium; reverse trajectories show formation from noise. Both schedules produce comparable trajectory quality with the same trained constraint field.

5 References

- Sohl-Dickstein et al. (2015). *Deep Unsupervised Learning using Nonequilibrium Thermodynamics.* ICML.
- Ho, Jain, & Abbeel (2020). *Denoising Diffusion Probabilistic Models.* NeurIPS.
- Song & Ermon (2019). *Generative Modeling by Estimating Gradients of the Data Distribution.* NeurIPS.
- Nichol & Dhariwal (2021). *Improved Denoising Diffusion Probabilistic Models.* arXiv.